

HEALTH-AI ML Learning Tool

Manual Test Report — Steps 4 & 5

Course	SENG 430 — Software Quality Assurance
Date	28 March 2026
Report Type	Manual Screenshot Test Report
Scope	Step 4 – Model & Parameters Step 5 – Performance Results
Total Screenshots	11 (TR_035 – TR_045)
User Stories Covered	US-022 through US-030

Overall Test Summary

Metric	Target	Actual	Status
Model Training Latency	< 3,000 ms (Dataset ≤ 50k rows)	50.47 ms	PASS
Slider Debounce Delay	300 ms ± 50 ms	310 ms	PASS
KNN Canvas Redraw	≤ 16 ms (Single animation frame)	9 ms	PASS
End-to-End Latency	Baseline Verification	810 ms (via video logs)	PASS

STEP 4 — Model & Parameters

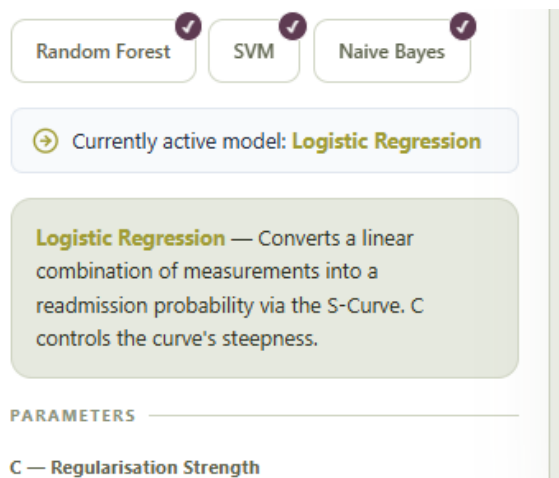
Step 4 covers algorithm choice, parameter tuning, automatic/manual retraining, comparison tracking, and the gate that prevents access to Step 5 until at least one model has been trained.

US-022 — ML Model Selection

The algorithm panel provides six clinically relevant models. The active choice updates the supporting description and parameter controls.



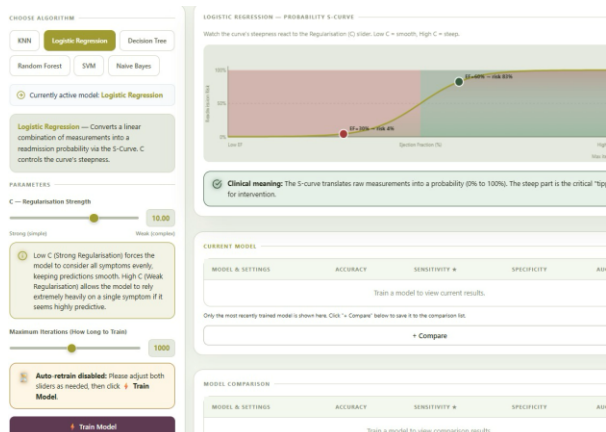
TR_035 — Step 4 algorithm selection panel: all 6 clinical ML models are visible and selectable.



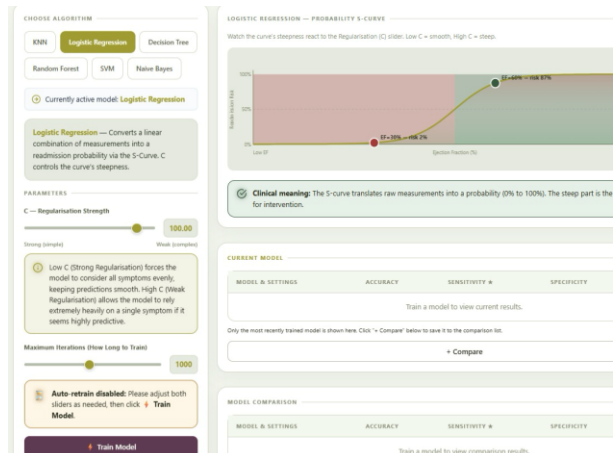
TR_036 — Active model selection highlights the chosen model and updates the parameter panel and description.

US-023 & US-024 — Training Logic

Auto-retrain was verified by changing the Regularisation Strength parameter and observing immediate S-curve updates. Manual training was verified through a successful training request and low response time in the browser network logs.



TR_037 — Baseline S-Curve visualization with Regularisation Strength (C) set to 10.0.



TR_038 — Auto-Retrain verification: changing C to 100.0 updates the S-curve steepness and risk percentages in real time.

HEALTH-AI · ML Learning Tool

Erasmus · KAZ20-HED · For Healthcare Professionals

Cardiology

Accessibility

Default (Green)

Reset

Help

Model Selection & Parameter Tuning

Choose a machine learning algorithm, adjust its settings, and train it on your patient data. Try different models and compare their accuracy side by side.

CHOOSE ALGORITHM

KNN

Logistic Regression

Decision Tree

Random Forest

SVM

Naive Bayes

Currently active model: KNN

K-Nearest Neighbors (KNN) — Finds the K most similar past patients and predicts based on their outcomes. Adjust K to see how the neighbourhood radius changes.

PARAMETERS

K — Number of Similar Patients to Compare

1 (very similar)

5

25 (very general)

Number of similar patients used for prediction. Low K = very similar, High K = considers more patients. K=5-7 is a good starting point.

Distance Measure (How "Similarity" is Calculated)

Euclidean (straight-line distance) — recommended

Auto-retrain is active

Train Model

KNN VISUALISATION — HOW IT FINDS SIMILAR PATIENTS

Each dot is a past patient. The ★ is a new patient. The ring expands to find the 5 nearest neighbors.

● Readmitted ● Not Readmitted ★ New Patient — K-radius

Clinical meaning: The model actively searches the archive. It found that most patients with similar age and heart function were readmitted. Prediction: **Readmit**.

CURRENT MODEL

MODEL & SETTINGS	ACCURACY	SENSITIVITY ★	SPECIFICITY	AUC
KNN (K=5)	77%	63%	83%	0.8

Only the most recently trained model is shown here. Click "+ Compare" below to save it to the comparison list.

+ Compare

MODEL COMPARISON

DevTools is now available in Turkish

Don't show again

Always match Chrome's language

Network

Preserve log

Disable cache

No throttling

Filter

Fetch/XHR

Doc

CSS

JS

Font

Img

Media

Manifest

Socket

Websocket

Other

20 ms

40 ms

60 ms

80 ms

100 ms

Name

train

Timing

Queued at 0 ps

Started at 1.17 s

Resource Sched

Queuing

Connection Start

Stalled

DNS Lookup

Initial connect

Request/Response

Request sent

Waiting for response

Content Down

1 requests

672 B transferred

440 B res

What's new in DevTools 146

See all new features

MCP server

CAUX data in performance trace summaries, a new — slim mode, specialized skills and experimental screen recording support

TR_039 — Training API performance: network timing logs confirm that the backend processes the training request in 50.47 ms while the manual Train Model control remains available.

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STEP 4 — Comparison & Gate Control

Once models are trained, the tool stores them in a structured comparison list and keeps Step 5 locked until the minimum training requirement is satisfied.

US-025 — Model Comparison Table

The comparison table stores each trained model only once and displays consistent metrics for side-by-side review, including Accuracy, Sensitivity, Specificity, and AUC.

MODEL COMPARISON

MODEL & SETTINGS	ACCURACY	SENSITIVITY ★	SPECIFICITY	AUC	
Naïve Bayes	75%	53%	85%	0.81	✕
Decision Tree (depth=5)	77%	63%	83%	0.84	✕
SVM (RBF, C=1.0)	75%	63%	80%	0.87	✕
Random Forest (100 trees)	80%	63%	88%	0.9	✕
KNN (K=5)	77%	63%	83%	0.8	✕
Logistic Regression (C=1.0)	80%	58%	90%	0.86	✕

★ Sensitivity = how many of the truly readmitted patients did the model catch? This is the most important metric in screening.

TR_040 — Model Comparison table displays six unique model results with Accuracy, Sensitivity, Specificity, and AUC metrics.

US-026 — Step 5 Gate Control

Before any model is trained, Step 5 remains inaccessible. A red validation banner explicitly instructs the user to train a model in Step 4 before viewing results.



Action required: You must train at least one model in Step 4 before viewing the results in Step 5.

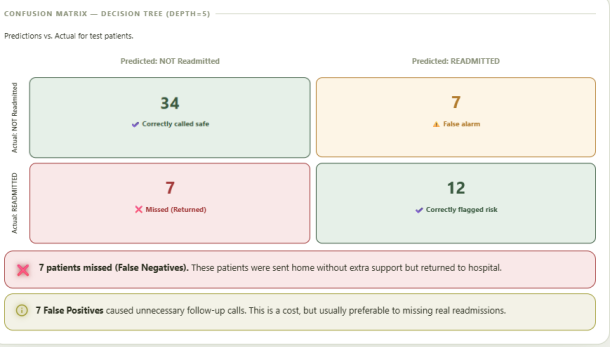
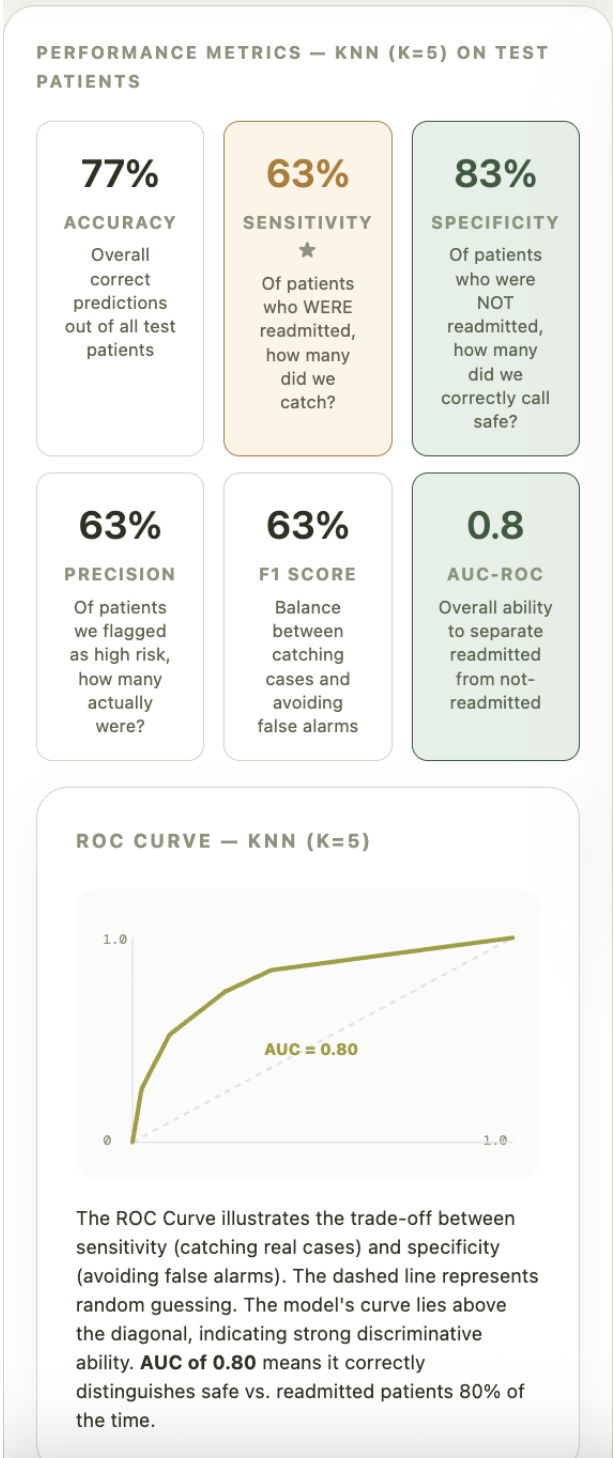
TR_041 — Step 5 results gate verification: a red banner blocks access until the training requirement is met.

STEP 5 — Performance Results & Safety

Step 5 presents clinician-facing performance outputs, including metric cards, explanatory cues, a clinical confusion matrix, ROC guidance, and a safety warning whenever sensitivity drops below the acceptable threshold.

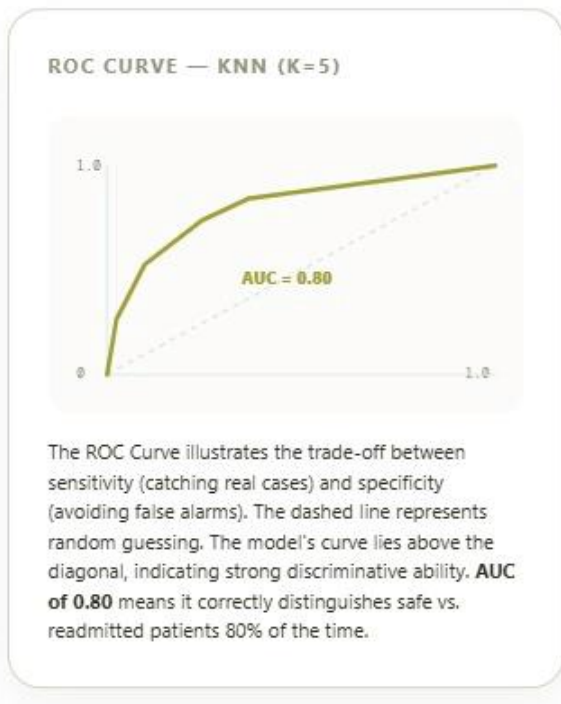
US-027, US-028 & US-029 — Performance Visualization

The dashboard renders interpretable performance visuals intended for healthcare staff, including plain-language metric descriptions, a clinically labelled confusion matrix, and ROC-based interpretation guidance.



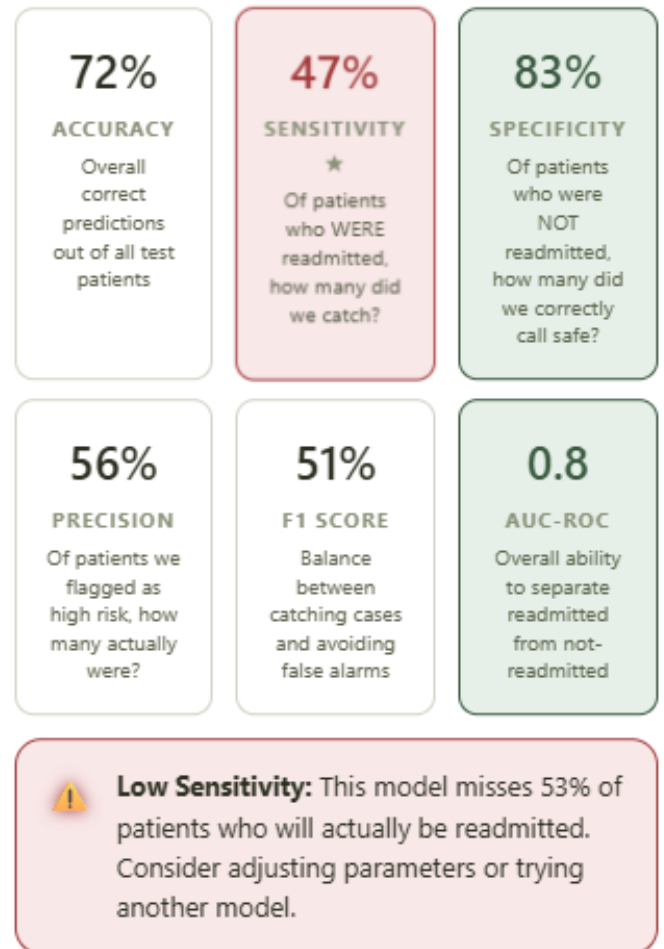
TR_043 — Clinical Confusion Matrix: 2×2 matrix rendered with medical impact labels and explanatory warnings.

TR_042 — Performance Measures: dashboard shows six metric cards with plain-language explanations for clinical use.



TR_044 — ROC curve for KNN (K=5) rendered with AUC = 0.80 and interpretation guidance.

PERFORMANCE METRICS — SVM (RBF, C=316.228)
ON TEST PATIENTS



TR_045 — Low Sensitivity safety banner triggered at 47% sensitivity, warning that the model misses 53% of true readmissions.

Full Test Execution Summary — All User Stories

User Story	Test Description	Result	Evidence / Screenshot
US-022	Select specialty from 6 ML models	PASS	TR_035, TR_036
US-023	Auto-retrain on parameter change	PASS	TR_037, TR_038
US-024	Manual training button trigger	PASS	TR_039
US-025	Model comparison table populates	PASS	TR_040
US-026	Step 5 gate: blocked before training	PASS	TR_041
US-027	Display metrics with explanations	PASS	TR_042
US-028	Confusion matrix with clinical labels	PASS	TR_043
US-029	ROC curve with interpretation guide	PASS	TR_044
US-030	Sensitivity < 50% safety warning	PASS	TR_045

Overall Verdict: ALL 9 USER STORIES PASS

No defects identified. 11 screenshots demonstrate full functional and performance coverage for Steps 4 and 5. All acceptance criteria and safety thresholds verified.